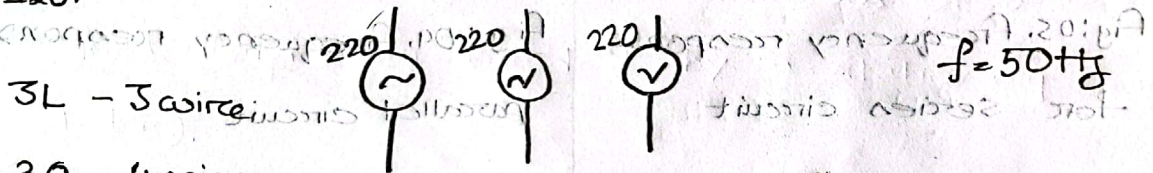


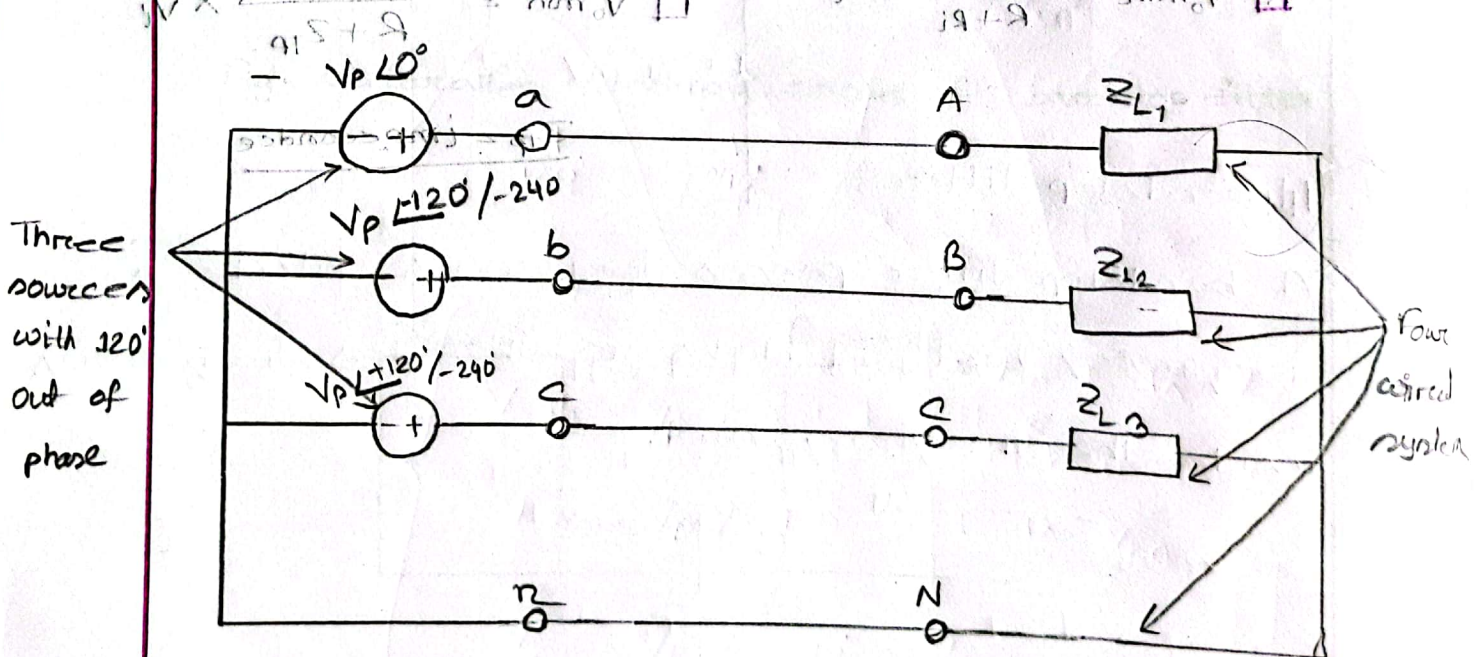
Three - Phase System

□ What is a three phase circuit (1)?

It is a system produced by a generator consisting of three sources having the same amplitude and frequency, but out of phase with each other by 120° .



3Q - 4 wire



□ What is a ~~three~~ phase circuit (2)?

Advantage:

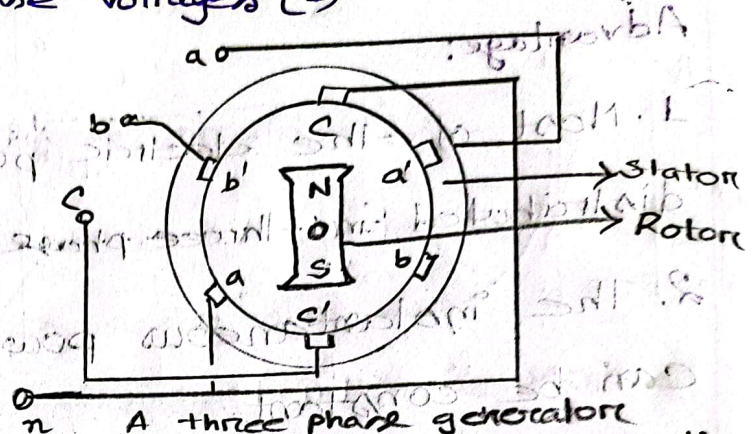
1. Most of the electric power is generated and distributed in three-phase
2. The instantaneous power in a three-phase system can be constant.
3. The amount of power the three-phase system is more economical than the single phase
4. In fact the amount of wire required for a three phase system is less than for an equivalent single phase system.

□ Balance three phase Voltages (1)

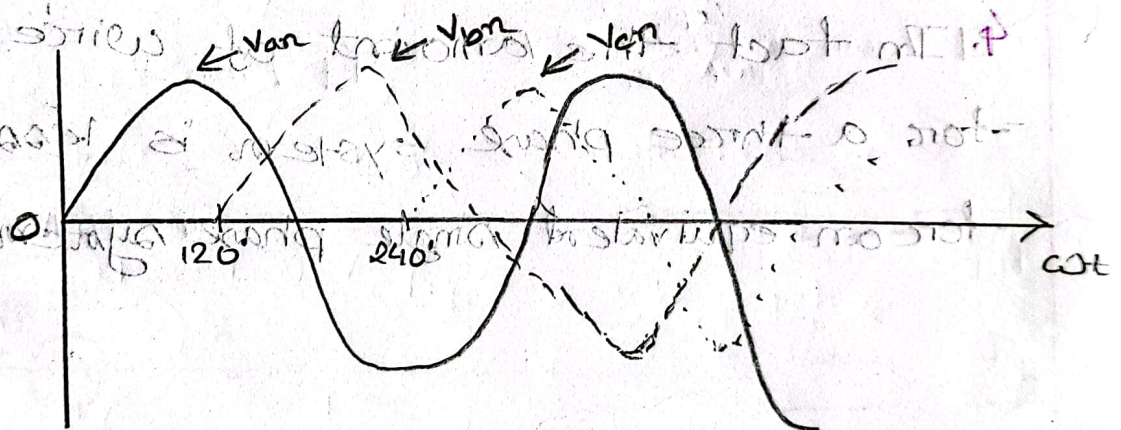
• 3- ϕ type $\rightarrow$

$\rightarrow$ Unbalanced

$\checkmark \rightarrow$ Balanced



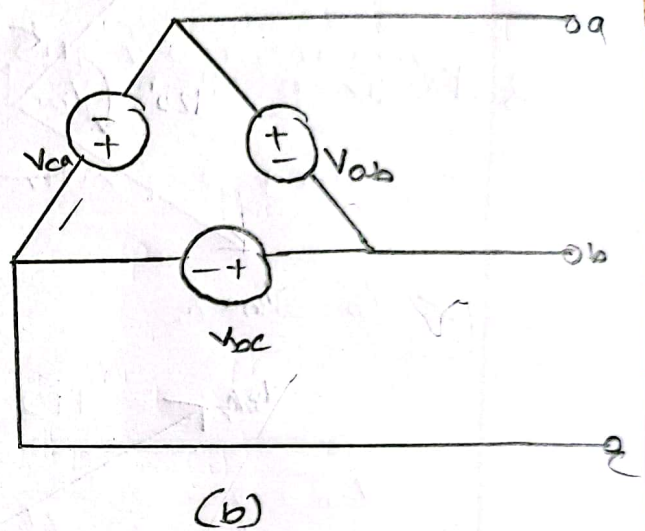
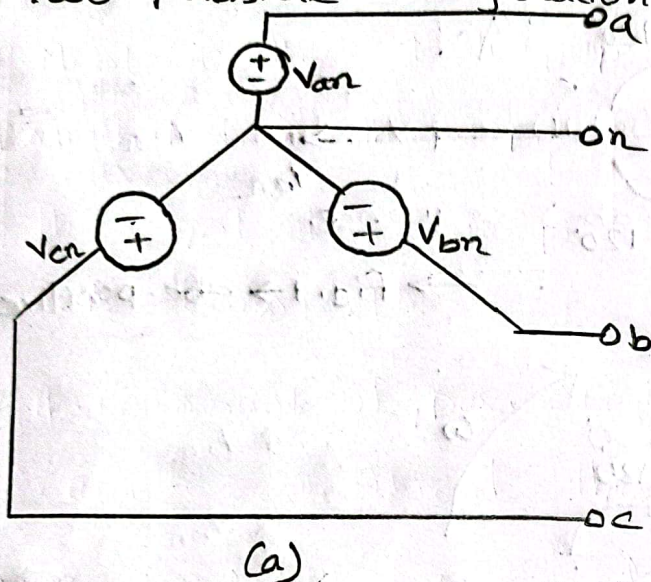
$\Rightarrow$ A three-phase generator consists of a rotating magnet (rotor) surrounded by a stationary winding (Stator).



The generated Voltage

□ Balance three phase voltages (2)

- Two possible configuration →



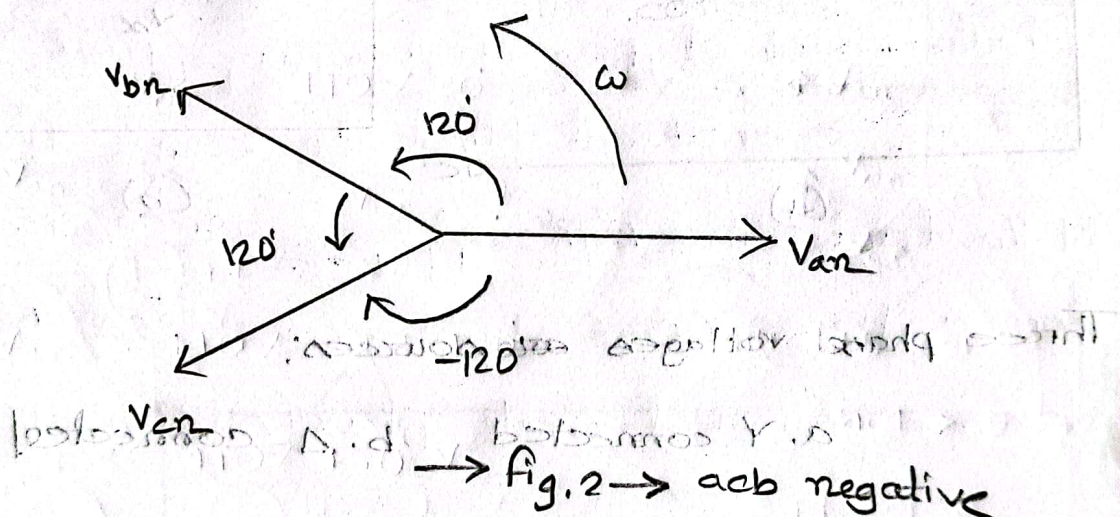
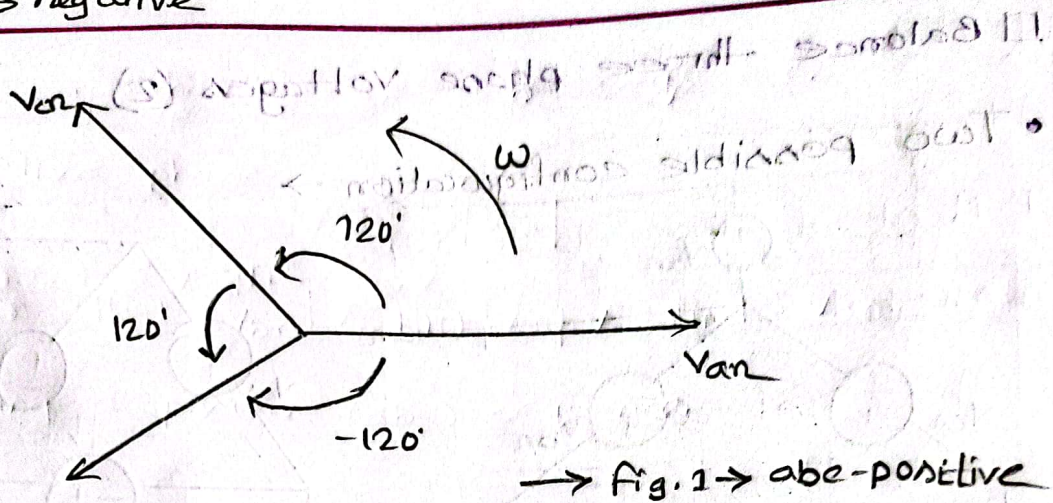
Three phase voltages are sources.

a. Y connected b. Δ - connected

□ Balance three phase voltages (3)

- Balanced phase voltages are equal in magnitude and are out of phase with other by 120° .
- The phase sequence is the time order in which the voltages pass through their respective maximum values.
- A Balanced load is one in which the phase impedances are equal in magnitude and in phase.

- phase sequence
- * $abc \rightarrow$ positive
- * $acb \rightarrow$ negative



$\left\{ \begin{array}{l} V_{an} = V_p \angle 0^\circ \\ V_{cn} = V_p \angle -120^\circ \\ V_{bn} = V_p \angle -240^\circ = V_p \angle +120^\circ \end{array} \right.$	• V_{an} leads V_{cn} - which in turn - leads V_{bn}
--	---

□ Balance three phase Voltages (4)

* Q.

Example: 1. Determine the phase sequence of the set of voltages.

$$V_{an} = 200 \cos(\omega t + 10^\circ)$$

$$V_{bn} = 200 \cos(\omega t - 230^\circ)$$

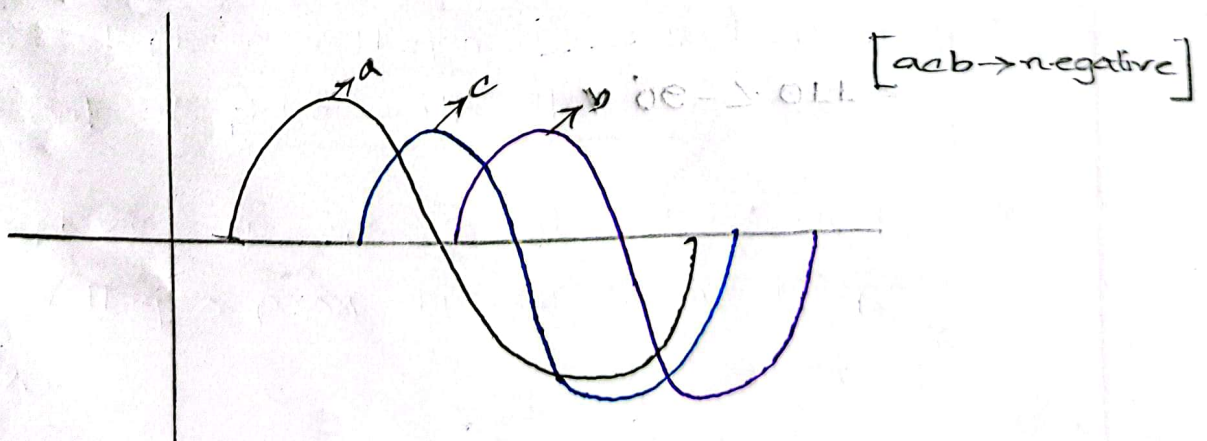
$$V_{cn} = 200 \cos(\omega t - 110^\circ)$$

⇒ The voltages can be expressed in phasor form as

$$V_{an} = 200 \angle 10^\circ \text{ V}$$

$$V_{bn} = 200 \angle -230^\circ \text{ V}$$

$$V_{cn} = 200 \angle -110^\circ \text{ V}$$



We notice that V_{an} leads V_{cn} by 120° , and V_{cn} in turn leads V_{bn} by 120°

Hence we have an acb sequence,

negative phase sequence.

Ans: $110 \angle 150^\circ \text{ V}$, $110 \angle -90^\circ \text{ V}$

Q. * Given that $V_{bn} = 110 \angle 30^\circ \text{ V}$, find V_{an} and V_{cn} assuming a positive (abc) sequence

$\Rightarrow$ V_b Given,

$$V_{bn} = 110 \angle 30^\circ$$

$$V_{an} = ?$$

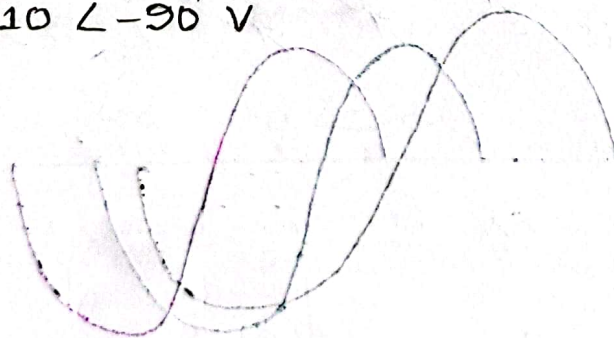
$$V_{cn} = ?$$

$$V_{an} = 110 \angle 30^\circ + 120^\circ$$

$$= 110 \angle 150^\circ \text{ V}$$

$$V_{cn} = 110 \angle 30^\circ - 120^\circ$$

$$= 110 \angle -90^\circ \text{ V}$$



□ Balance three phase connection (1)

• Four Possible connections →

1. Y-Y connection

2. Y-Δ

3. Δ-Δ

4. Δ-Y

* Y-Y connection →

source is Y connected

load is Y connected

* Y-Δ connection →

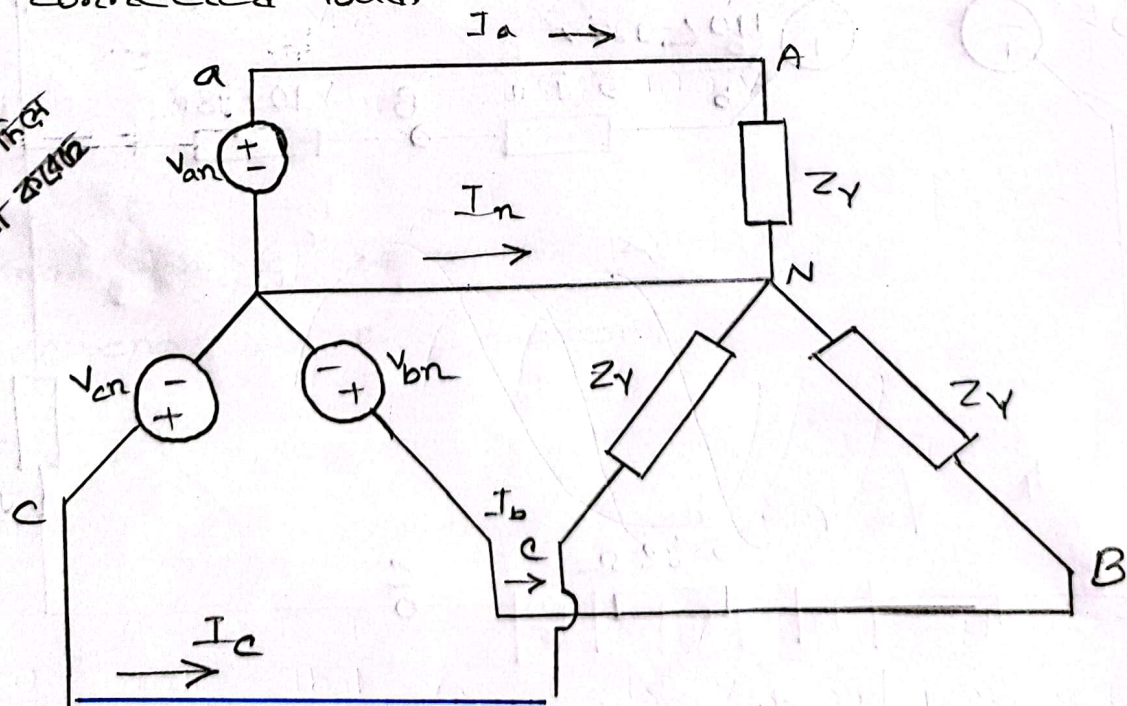
source is Y connected

load is Δ connected

→ यदि 2 टिप 2 टिप,

□ Balance three phase connection (2)

• A balanced Y-Y system is a three-phase system with a balanced y-connected source and a balanced y-connected load.



* Load is 2 first
→ यदि 2 टिप 2 टिप

* $V_L = \sqrt{3}(V_p)$, where

$$V_p = |V_{an}| = |V_{bn}| = |V_{cn}|$$

$$V_L = |V_{ab}| = |V_{bc}| = |V_{ca}|$$

Q * Calculate the line currents in the three-wire Y-Y system shown below:

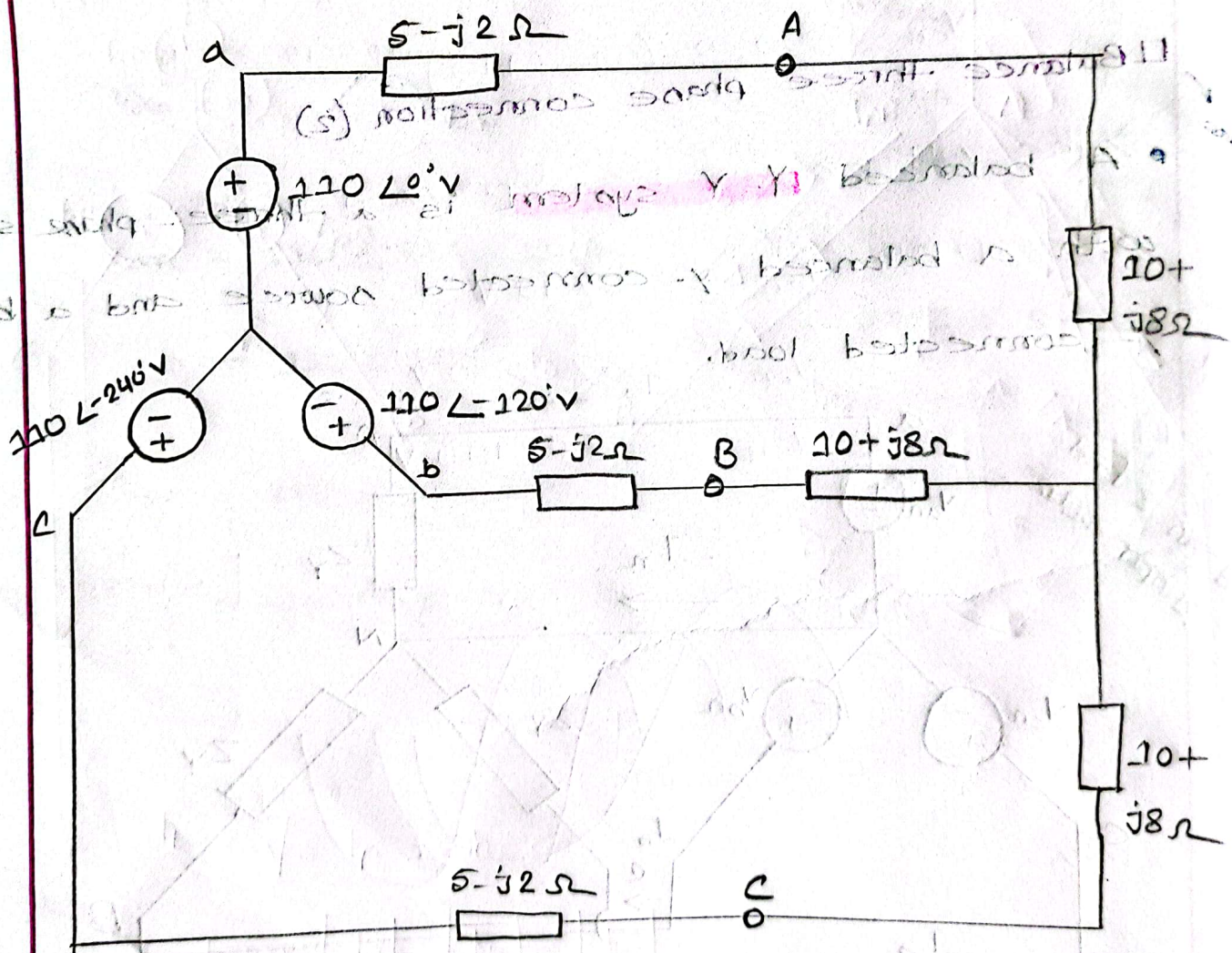


Fig. 1.

⇒ The three phase circuit in fig. 1 is balanced, we may replace it with its single phase equivalent circuit such as in fig. 2. We obtain I_a from the single-phase analysis as

$$I_a = \frac{V_{an}}{Z_Y}$$

[যেকোনো স্টা-পয়েন্ট
করেছি কারণ সব

Where, $Z_Y = (5 - j2) + (10 + j8)$ Same জিরিটে
[আছে]

$$= 15 + j6$$

$$= 16.155 \angle 21.8^\circ$$

Hence,

$$I_a = \frac{110 \angle 0^\circ}{16.155 \angle 21.8^\circ}$$

[যখন load Y connected

তখন $I_a = I_b = I_c$

বাঁকি গুলোও same তাই

আমাদা করে এর করে না]

$$= 6.81 \angle -21.8^\circ \text{ A}$$

Since the source voltages in fig. 1 are in positive sequence, the line currents are also in positive

4

$$I_b = I_a \angle -120^\circ = 6.81 \angle -21.8 - 120^\circ = 6.81 \angle -141.8^\circ \text{ A}$$

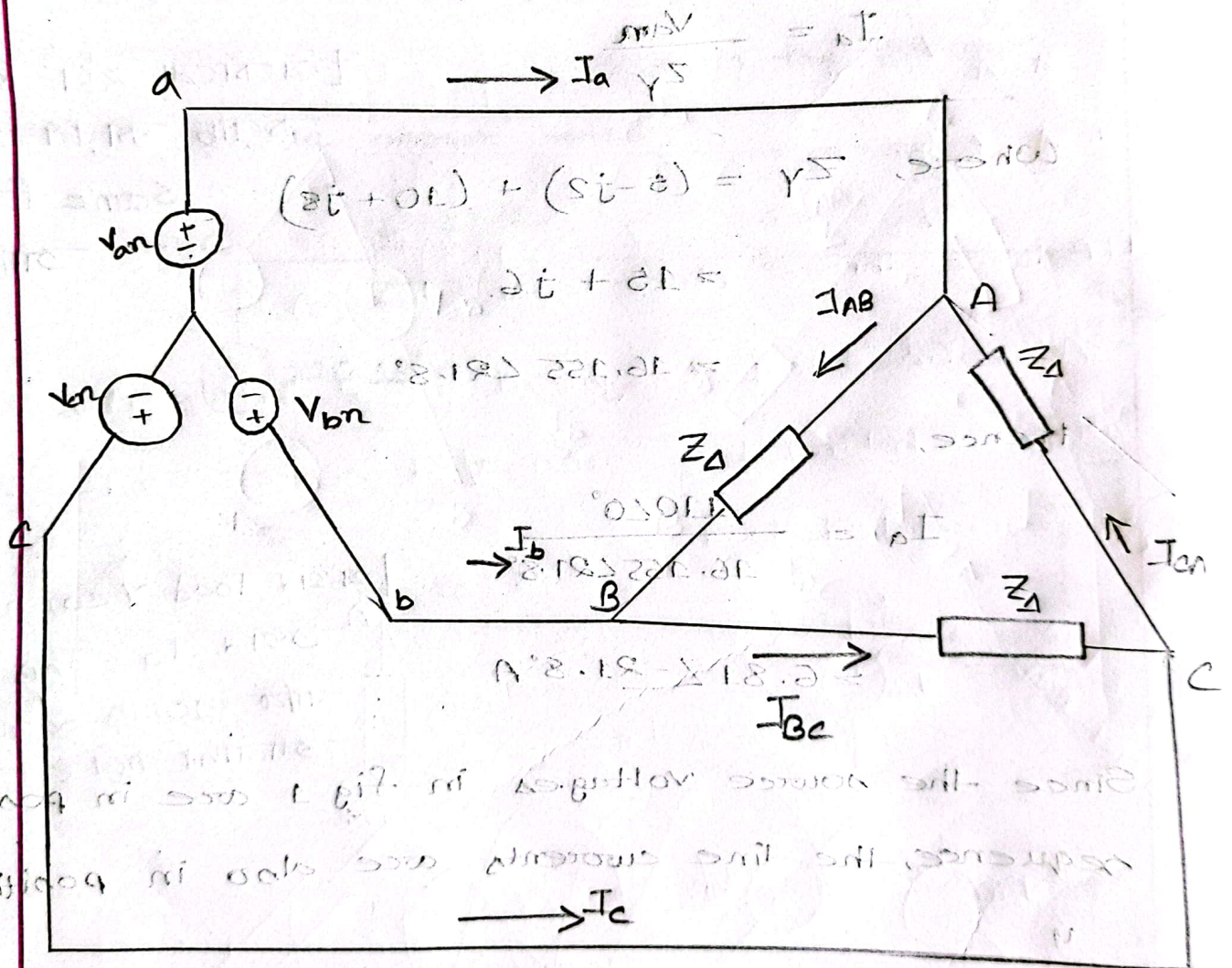
$$I_c = I_a \angle -240^\circ = 6.81 \angle -21.8 - 240^\circ = 6.81 \angle -261.8^\circ \text{ A} = 6.81 \angle 98.2^\circ \text{ A}$$

$$= 6.81 \angle -21.8 - 240^\circ$$

$$= 6.81 \angle -21.8 + 120^\circ$$

□ Balance three phase connection (4)

- A balanced **Y-Δ system** is a three-phase system with a balanced y-connected source and a balanced Δ-connected load.



$$I_L = \sqrt{3} I_p \text{ per phase}$$

$$I_L = |I_a| = |I_b| = |I_c|$$

$$I_p = |I_{AB}| = |I_{BC}| = |I_{CA}|$$

Q * A balanced abc-sequence Y-connected source with $(V_{an} = 100 \angle 10^\circ)$ connected to a Δ -connected load $(8 + j4) \Omega$ per phase. Calculate the phase and line currents.

$\Rightarrow$ Using single phase analysis,

$$\begin{aligned} I_a &= \frac{V_a}{Z_{\Delta}/3} \\ &= \frac{100 \angle 10^\circ}{2.981 \angle 26.57^\circ} \\ &= 33.54 \angle -16.57^\circ \text{ A} \end{aligned}$$

Other line currents are obtained using the abc phase sequence

$$\begin{aligned} I_b &= 33.54 \angle -16.57^\circ - 120^\circ \\ &= 33.54 \angle -136.57^\circ \text{ A} \end{aligned}$$

$$\begin{aligned} I_c &= 33.54 \angle -16.57^\circ - 240^\circ \\ &= 33.54 \angle -256.57^\circ \text{ A} \end{aligned}$$

$$I_L = \sqrt{3} I_p \angle -30^\circ$$

$$\therefore I_p = \frac{I_L}{\sqrt{3}} \angle 30^\circ \text{ [Angle eqn at opposite to different sign change]}$$

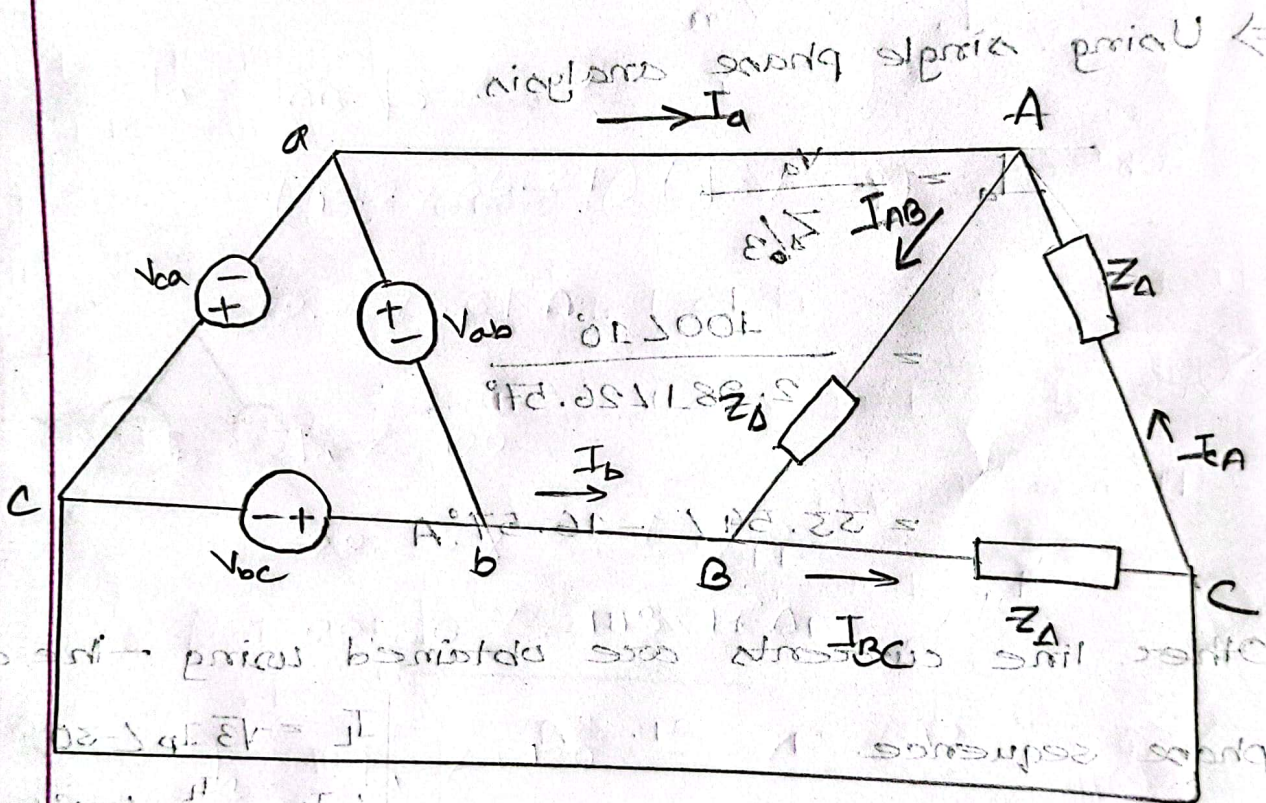
$$\begin{aligned} \therefore I_{AB} &= \frac{I_a}{\sqrt{3}} \angle 30^\circ \\ &= \frac{33.54 \angle -16.57^\circ}{\sqrt{3}} + \angle 30^\circ \\ &= 19.37 \angle 13.43^\circ \text{ A} \end{aligned}$$

$$\begin{aligned} \therefore I_{BC} &= 19.37 \angle 13.43^\circ - 120^\circ \\ &= 19.37 \angle -106.57^\circ \text{ A} \end{aligned}$$

$$\begin{aligned} \therefore I_{CA} &= 19.37 \angle 13.43^\circ - 240^\circ \\ &= 19.37 \angle -226.57^\circ \text{ A} \end{aligned}$$

□ Balance Three phase connection (b)

- A balanced Δ - Δ system is a three phase system with a balanced Δ -connected source and a balanced Δ -connected load.



Q * A balanced Δ -connected load having an impedance $20 - j15 \Omega$ is connected to a Δ -connected positive sequence generator having ($V_{ab} = 330 \angle 0^\circ \text{ V}$).

Calculate the phase currents of the load, and the line currents.

$\Rightarrow$ The phase (currents) (I_{AB}, I_{BC}, I_{CA}) =

$$\begin{aligned} I_{AB} &= \frac{V_{ab}}{Z_{\Delta}} \\ &= \frac{330 \angle 0^\circ}{20 - j15} \\ &= 13.2 \angle 36.87^\circ \text{ A} \quad [\text{Using Calculator}] \end{aligned}$$

$$\therefore I_{BC} = 13.2 \angle 36.87^\circ - 120^\circ \text{ A}$$

$$= 13.2 \angle -83.13^\circ \text{ A}$$

$$\therefore I_{CA} = 13.2 \angle 36.87^\circ - 240^\circ \text{ A}$$

$$= 13.2 \angle -203.13^\circ \text{ A}$$

$$= 13.2 \angle 36.87^\circ + 120^\circ \text{ A}$$

$$= 13.2 \angle 156.87^\circ \text{ A}$$

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For a delta load, the line current always lags
the corresponding phase current by 30° and has
a magnitude $\sqrt{3}$ times that of the phase current.
Hence, the line currents are

$$I_a = I_{AB} \sqrt{3} \angle -30^\circ$$

$$= (13.2 \angle 36.87^\circ) (\sqrt{3} \angle -30^\circ)$$

$$= 22.86 \angle 6.87^\circ \text{ A}$$

$$I_b = I_a \angle -120^\circ$$

$$= 22.86 \angle 6.87^\circ - 120^\circ$$

$$= 22.86 \angle -113.13^\circ \text{ A}$$

$$I_c = I_a \angle +120^\circ \rightarrow -240^\circ \text{ or } 120^\circ$$

$$= 22.86 \angle 6.87^\circ + 120^\circ$$

$$= 22.86 \angle 126.87^\circ \text{ A}$$

$$\begin{cases} I_a = 22.86 \angle 6.87^\circ \text{ A} \\ I_b = 22.86 \angle -113.13^\circ \text{ A} \\ I_c = 22.86 \angle 126.87^\circ \text{ A} \end{cases}$$

$$\begin{cases} I_a = 22.86 \angle 6.87^\circ \text{ A} \\ I_b = 22.86 \angle -113.13^\circ \text{ A} \\ I_c = 22.86 \angle 126.87^\circ \text{ A} \end{cases}$$

□ Balance three phase - connection (8)

- A balanced **Δ -Y system** is a three-phase system with a balanced y-connected source and a balanced y-connected load.

